

KHUTAG-UNDUR, Mongolia

WMO: **442320**

Lat: **49.3936N** Lon: **102.7092E** Elev: **938** StdP: **90.55** Time Zone: **8.00 (E08)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-37.6	-34.5	-39.8	0.1	-37.5	-36.7	0.1	-34.3	8.6	-14.5	7.3	-14.2	0.0	210	0.622	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	12.1	30.9	18.3	28.8	17.3	26.9	16.9	20.5	27.5	19.3	25.7	18.2	24.2	2.5	90	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	18.4	14.9	23.5	17.2	13.8	22.0	16.1	12.8	20.9	63.9	27.2	59.4	25.9	55.8	24.1	27.8	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	8.4	7.3	6.5	DB	-38.5	35.0	3.8	2.1	-41.2	36.5	-43.5	37.8	-45.6	38.9	-48.4	40.5	(4)
(5)				WB	-38.4	22.3	3.9	2.0	-41.2	23.7	-43.5	24.8	-45.7	25.9	-48.6	27.3	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		0.7	-23.6	-17.3	-5.0	5.1	11.5	17.5	19.6	17.3	10.8	1.8	-10.7	-20.2	(6)
(7)		DBStd		15.87	6.19	6.81	6.83	4.78	4.62	3.81	3.15	3.16	4.15	4.74	7.15	7.07	(7)
(8)		HDD10.0		4318	1042	765	467	156	38	2	0	1	39	253	620	936	(8)
(9)		HDD18.3		6571	1300	998	724	396	214	59	22	57	226	511	870	1194	(9)
(10)		CDD10.0		911	0	0	0	10	86	225	299	227	64	1	0	0	(10)
(11)		CDD18.3		123	0	0	0	0	3	33	62	24	1	0	0	0	(11)
(12)		CDH23.3		1541	0	0	0	3	138	479	634	259	27	0	0	0	(12)
(13)		CDH26.7		471	0	0	0	0	33	156	212	67	2	0	0	0	(13)
(14)	Wind	WSAvg		1.9	0.8	1.2	2.3	2.9	2.8	2.5	2.3	2.1	2.2	1.9	1.3	1.0	(14)
(15)	Precipitation	PrecAvg		141	20	19	12	7	9	4	4	4	12	6	12	14	(15)
(16)		PrecMax		401	272	100	116	30	34	34	25	35	60	27	65	252	(16)
(17)		PrecMin		22	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd		86	41	32	20	8	10	7	6	8	15	7	17	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		-3.7	2.4	15.5	23.6	30.0	33.6	34.0	31.9	26.3	18.7	9.2	-1.5	(19)
(20)			MCWB		-5.7	-1.8	6.3	10.9	14.5	18.0	20.5	19.5	15.5	9.6	3.9	-4.6	(20)
(21)		2%	DB		-8.4	-0.9	11.1	19.8	27.0	30.5	31.2	28.2	23.6	15.4	5.5	-4.1	(21)
(22)			MCWB		-10.1	-4.1	3.8	9.3	13.7	17.3	19.4	17.9	14.0	7.9	1.1	-6.6	(22)
(23)		5%	DB		-11.3	-3.9	8.1	17.3	24.4	28.3	29.0	26.0	21.2	12.7	2.7	-6.6	(23)
(24)			MCWB		-12.4	-6.3	2.2	7.9	12.7	16.6	18.5	17.4	13.0	6.2	-0.8	-8.5	(24)
(25)		10%	DB		-14.2	-6.8	5.3	14.8	21.8	26.1	26.9	24.1	18.8	10.1	-0.3	-10.0	(25)
(26)			MCWB		-15.1	-8.6	0.4	6.6	11.5	15.8	17.9	16.8	11.8	4.7	-2.6	-11.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		-5.7	-1.5	7.0	12.1	16.9	20.6	23.1	21.4	17.1	10.2	3.8	-4.4	(27)
(28)			MCDB		-3.7	2.0	14.9	20.6	26.6	30.6	31.5	28.3	23.2	17.5	8.7	-1.6	(28)
(29)		2%	WB		-10.0	-4.0	4.2	9.7	15.0	19.0	21.3	19.4	15.1	8.2	1.3	-6.5	(29)
(30)			MCDB		-8.5	-1.1	10.5	18.5	24.8	27.7	28.2	25.6	21.9	14.7	5.1	-4.2	(30)
(31)		5%	WB		-12.4	-6.2	2.3	8.3	13.3	17.9	20.0	18.4	13.8	6.5	-0.5	-8.5	(31)
(32)			MCDB		-11.3	-4.1	7.7	16.3	22.6	25.3	26.2	24.0	20.0	12.3	2.5	-6.6	(32)
(33)		10%	WB		-15.1	-8.5	0.6	6.8	11.8	16.8	18.9	17.3	12.4	4.9	-2.7	-11.2	(33)
(34)			MCDB		-14.3	-6.8	4.9	14.3	20.6	23.6	24.5	22.8	17.8	9.6	-0.2	-10.1	(34)
(35)	Mean Daily Temperature Range		MDBR		11.1	14.7	15.0	14.3	14.9	13.7	12.1	12.3	14.2	14.2	11.5	9.8	(35)
(36)		5% DB	MCDBR		13.5	16.8	17.0	18.6	19.9	17.9	16.2	16.0	17.9	18.2	14.2	11.8	(36)
(37)			MCWBR		12.0	13.9	10.6	10.5	9.8	7.6	6.7	7.1	8.8	11.1	10.3	10.0	(37)
(38)		5% WB	MCDBR		13.4	16.6	16.4	17.4	18.1	15.3	13.6	13.6	16.3	17.6	13.5	11.7	(38)
(39)			MCWBR		12.0	13.9	10.5	10.5	9.9	7.6	6.6	6.9	8.4	11.0	10.0	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.248	0.272	0.320	0.362	0.393	0.395	0.384	0.359	0.315	0.294	0.270	0.251	(40)
(41)		taud			2.322	2.286	2.241	2.247	2.193	2.294	2.376	2.424	2.488	2.468	2.368	2.292	(41)
(42)		Ebn at Noon			822	881	886	882	865	864	869	878	891	850	784	757	(42)
(43)		Edh at Noon			66	89	112	125	138	126	115	104	88	74	64	60	(43)
(44)	All-Sky Solar Radiation	RadAvg			1.68	2.82	4.25	5.34	6.30	6.39	6.05	5.23	4.38	2.92	1.77	1.31	(44)
(45)		RadStd			0.08	0.15	0.16	0.26	0.32	0.44	0.32	0.30	0.20	0.11	0.08	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.72	-1.01	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	-1.82	N/A	N/A	-0.73	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page